

FLAGLEAF

Pilot v2 — from weed photos to field plans

Repositioning the pilot around the product we're actually building · AO «H3K» · 2026

1. The shift

Pilot v1 asked: "*will agronomists photograph field problems?*" — which bootstrapped weed/disease recognition. That capability is real, but it is a **feature, not a product**: general vision models (Gemini etc.) are catching up monthly, so "better recognition" is not a defensible business. And the data premise breaks as the season cleans the fields — there are fewer problems to shoot (a cleanliness itself achieved by **over-application of chemicals**).

Pilot v2 repositions around the real product: turning a field's state into an **efficient chemical & machine plan** — and updating that plan as new observations arrive.

2. The product thesis (why this is a product, not a feature)

sense the field → produce / update a plan → execute via machines & chemicals (CropWise) → observe the outcome → revise

The moat is not recognition — it is this **closed loop on a specific field over time**. A general AI has none of: the field's treatment history, the CropWise execution link, the locally **registered-product constraints** (Госкаталог), local agronomy, or the longitudinal outcome data. Recognition is one *sensor* feeding the loop; the **plan is the product**.

3. The new data unit: a scouting pass, not a hero photo

A scouting pass tracks the **weed spectrum and growth stage across the field over the season** — which families are present, at what stage, how close to the economic threshold (ЭПВ). That is the data the plan needs to call the **right timing, the right registered product, and the right dose**.

We change the unit of collection from "*a photo of a weed*" to "*a scouting pass that covers the field*," repeated through the season. It always exists (solving "nothing to shoot") and it feeds a decision.

Corrected by the chief agronomist (and this is the pilot working): for spring crops like soy the field is sprayed **whole, not by zones** — weeds emerge in waves from a field-wide seed bank, most herbicides are **contact** (no soil residual), and there are no variable-rate sprayers. So the saving is **not less area**; it is the better **decision** — correct ЭПВ **timing** (one pass instead of two), the right **product** for the actual spectrum, and the right **dose** (no crop injury from over-dosing).

4. How today's work fits (nothing is thrown away)

Current work	Becomes, in Pilot v2
 Photo shooting & recognition	Perception layer — ground-truth that calibrates field-state sensing and feeds the plan. Reframed from "hunt a problem" to "scout the field".
 CropWise operation logging	Execution & history layer — what was applied, where, by which machine. Already built; it is the loop's "execute + memory".



**NDVI watch /
satellite**

Field-scale sensing (free tier) — coarse anomaly signal today; the drone layer comes next.



**Grounded assistant
+ product catalog +
literature**

Decision layer — the substrate the new **field-plan generator** reasons over.

The pilot doesn't restart — it **recomposes** what exists into the plan loop.

5. New success metric

Replace "15–30 photos / week" with: **a better decision that costs less chemistry** — the right ЭПВ timing (fewer passes), the right registered product for the spectrum, and the right dose (no crop injury) — measured against what was actually applied, on 1–2 pilot fields, plus the chief agronomist's judgment that the plan matches how he really works. Measurable **now** from the CropWise spray records.

6. Roadmap

NOW

Field-plan generator on existing data + scouting-pass capture. Quantify the plan-vs-blanket saving on 1–2 fields.

NEXT

One contracted drone flight on one field as a field-scale sensing proof (manual analysis — no pipeline yet).

LATER

Close the loop: versioned per-field plans that update on each pass, with measured efficacy of what was done.

7. What we need to run it

- Pick **1–2 pilot fields** to run the full loop on.
- Agronomist runs a **scouting pass** (covers the field, clean areas included) instead of chasing problems.
- We generate a **treatment plan** and compare it to what was actually sprayed → the saving number.
- Book **one drone flight** on one field to validate field-scale sensing.